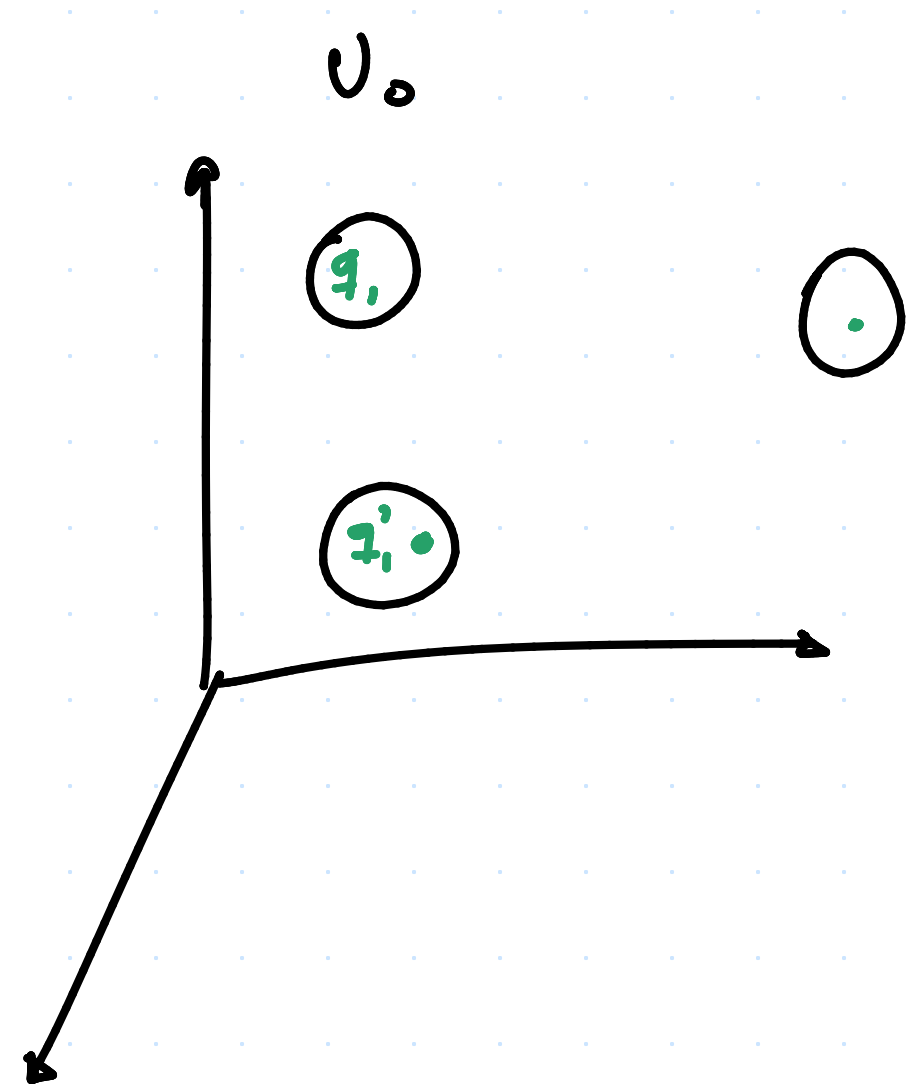


"Método de imágenes"

Unit 1: Electrostatics

Method of images



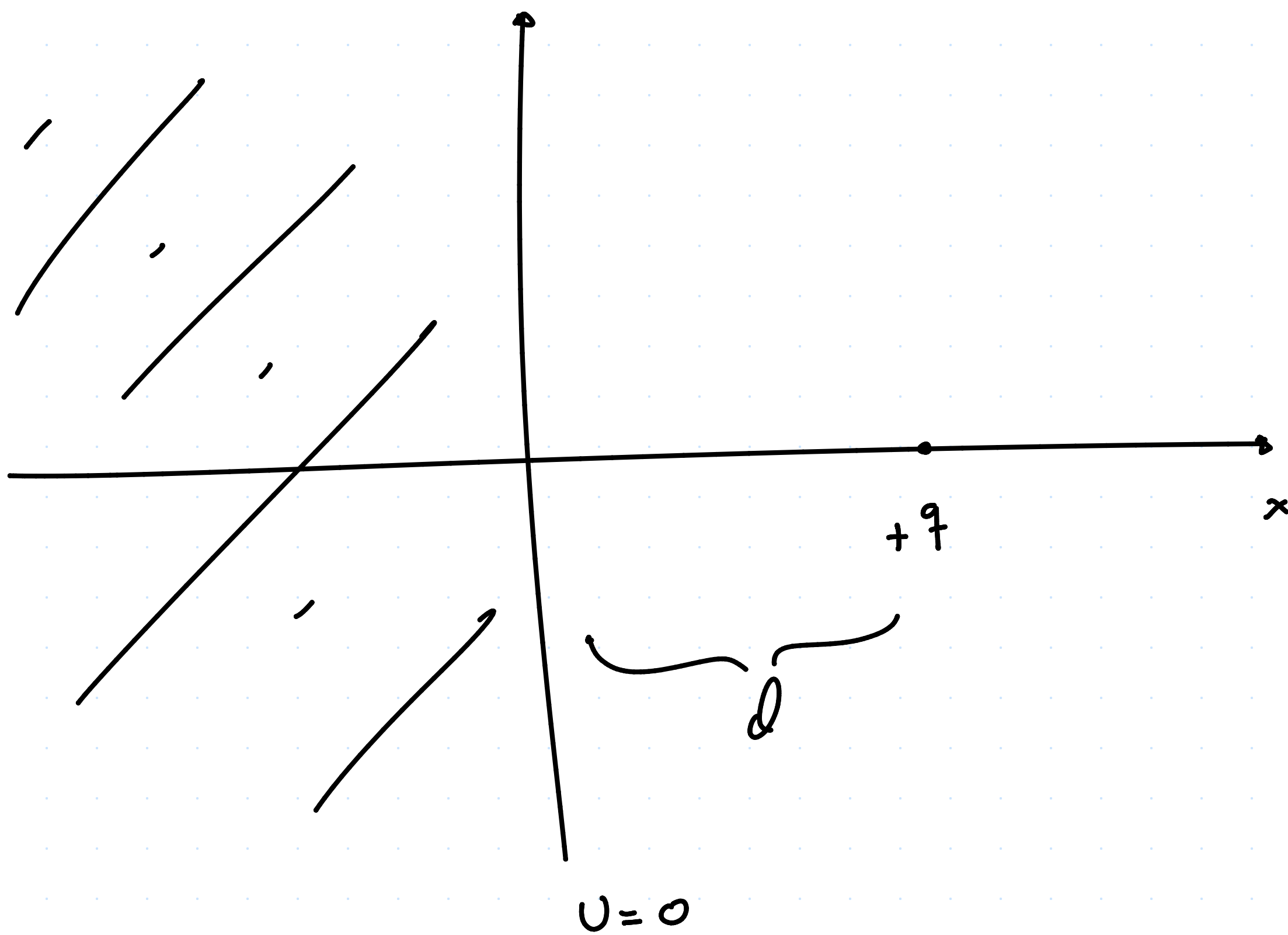
Allows us to solve some problems in electrostatics without explicitly solve the Laplace equation.

— it assumes that the potential can be written as

$$U = U_1 + \underbrace{\frac{1}{4\pi\epsilon_0} \int_S \frac{\sigma(\vec{r}') da}{|\vec{r} - \vec{r}'|}}_{\text{Contribution of conductor surface}}$$

we want to obtain the solution of the electrostatic problem by defining image charges that account for the "second" contribution under the same boundary conditions of the problem:

Example:



Tenemos

$$U_1 = \frac{1}{4\pi\epsilon_0} \frac{q}{\sqrt{(x-d)^2 + y^2 + z^2}}$$

La solución

serán: potencial de carga inducida

$$U = U_1 + \frac{1}{4\pi\epsilon_0} \int_{x=0} \frac{\sigma(\vec{r}') da'}{|\vec{r} - \vec{r}'|}$$

El método de imagen se basa en hallar una carga imagen que en realidad no existe, pero nos da el efecto...

The general recipe stands for...

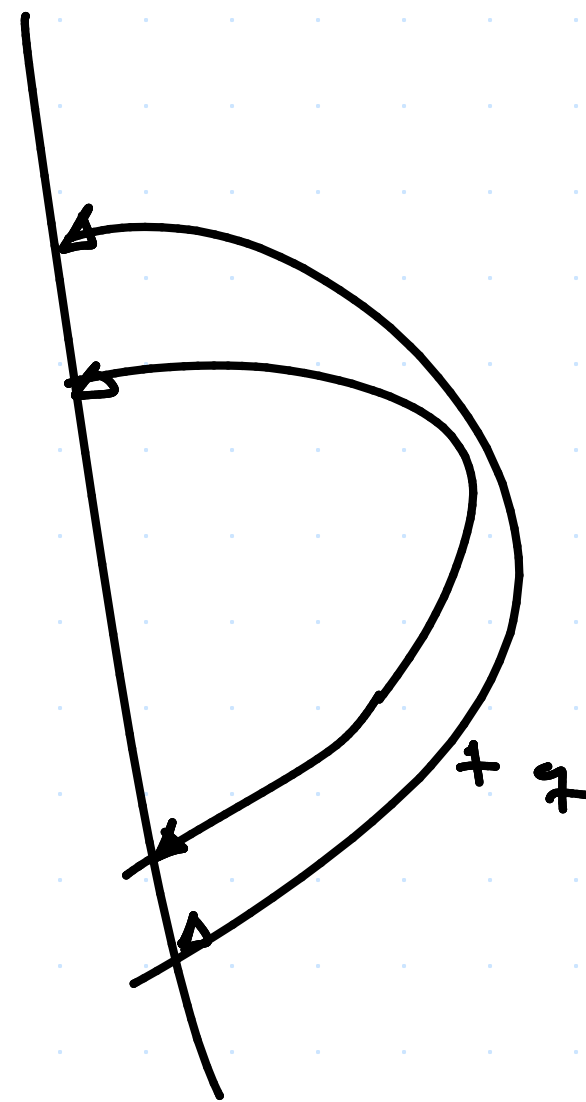
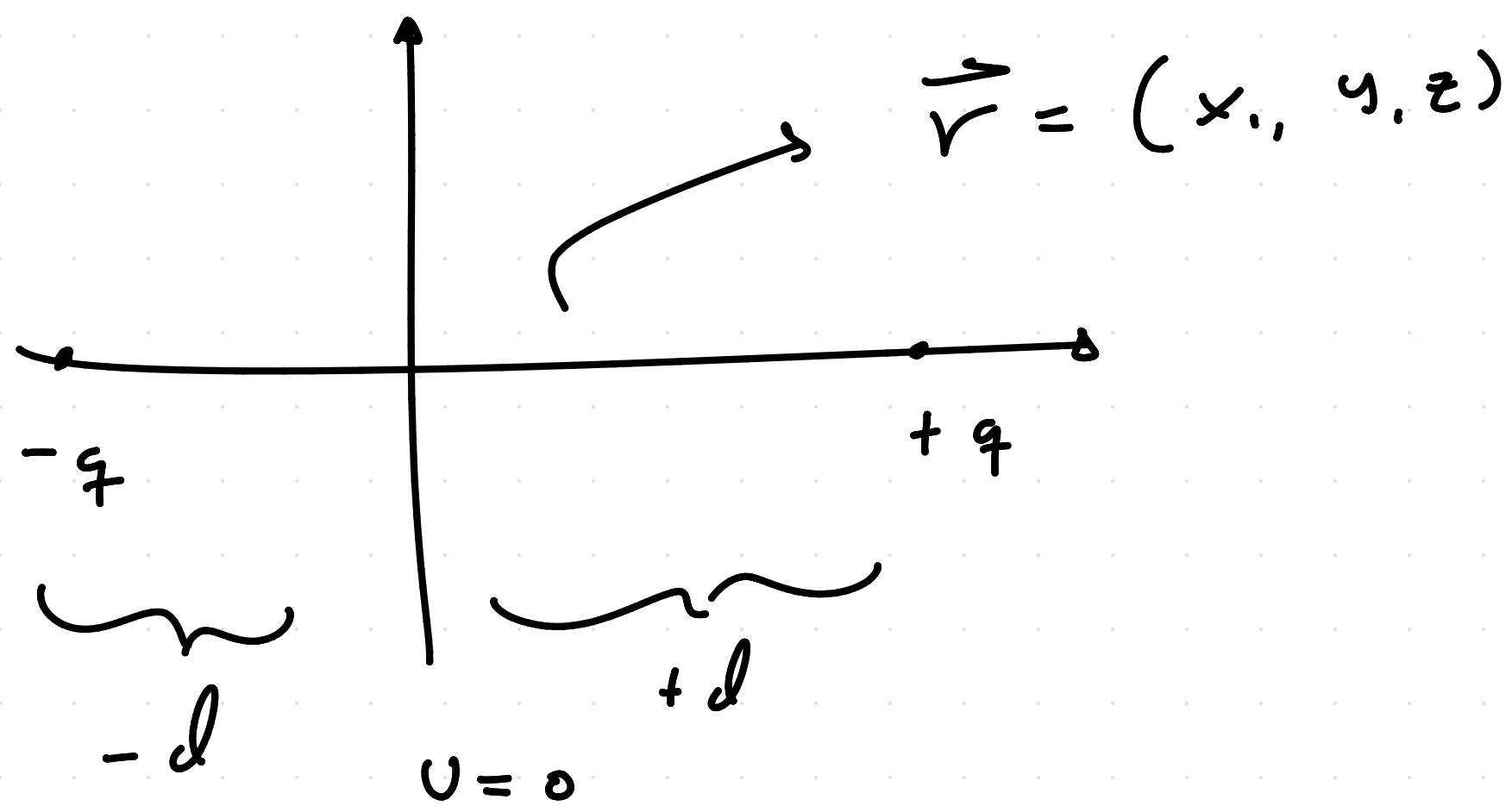


¿Dónde podríamos calcular una carga imagen?

$Q =$ una carga situada en el 2º cuadrante del plano colineal

a la carga positiva. (pág 4/4 Reitz ver... español)

pág 57 Reitz ver. inglés



Potencial debido a las dos contribuciones:

$$U(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{\sqrt{(x-d)^2 + y^2 + z^2}} - \frac{1}{4\pi\epsilon_0} \frac{q}{\sqrt{(x-(-d))^2 + y^2 + z^2}}$$

Ejercicio: Queremos hallar
por demostración

$$Q_{TS} = \int_{x=0} \sigma(\vec{r}) da' = -q$$

Gauss' law

$$EA = \frac{\sigma A}{\epsilon_0}$$

$\Rightarrow$

$$\sigma = \epsilon_0 E$$

$$1) E_x = - \left. \frac{\partial U}{\partial x} \right|_{x=0}$$

$$2) \sigma(\vec{r}) = \epsilon_0 E_x$$

3) Integral on the plane

